Humidification

Final Lab Report
CHE382: Chemical Engineering Laboratory 2

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TABLE OF CONTENTS

Executive Summary

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Where's the humidity data?

In this experiment mass and heat transfer are observed inside the wetted column by measuring experimental data in order to calculate heat and mass transfer coefficients. Experimental data is compared to the theoretical values calculated with Sherwood number correlation. The method used is humidification, as a simple representation of simultaneous heat and mass transfer inside the wetted wall column that is made of 2-inch diameter and 4 feet long Plexiglass. Air is transported from the bottom of the column in a countercurrent direction to the water that is flowing from the top of the column, with thermocouples placed accordingly to measure the temperatures inside the apparatus.

The system is initially tested without water to measure the humidity and temperature of dry air by using an electronic hygrometer. The resulting relative humidity of dry air is determined to be approximately 7%, and 0.0015 kg/m^3 of absolute humidity.
Water is then pumped into the system and the data is collected for different water flow rates, heated and unheated air. The water flow rates used are 3 SCFM, 6 SCFM and 9 SCFM. Different air temperatures used are 22.22 °C for no heat, 51 °C for medium heat, and 58, 60 and 65 °C for high heat.

The experimental results for mass and heat transfer coefficients agreed well with the Sherwood correlation at lower flowrates (Re<4000) but tended to deviate at higher flowrates (Re>4000). Due to experimental errors, limited conclusions may be made about the suitability of using Sherwood correlations for humidification columns. However, the following conclusions can be made on the trends of the experimental heat and mass transfer coefficients: the mass transfer coefficient increases as the flowrate increases and the heat transfer coefficient tended to decrease as the temperature of the air increased.

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You should present the trends in this experiment. What are the other parameters change? Such as the dimensionless groups and the NtG.

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Introduction

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Mass and heat transfer occur simultaneously in all types of processes. When the mass of a component is transferred during a phase transition, or from one liquid to another liquid or one gas to another gas, energy is also transferred in the form of heat. The interfacial temperature will adjust itself until a steady state rate of heat transfer is reached. Certain processes rely more on the significance of heat transfer than other processes, in which heat is of minor importance. In gas to liquid and vice versa transitions, the latent heat of condensation and evaporation, respectively, is involved in these conversions. A net transfer of mass occurs, and the heat transfer rate limits the rate at which this mass is transferred. However, in other operations such as distillation, liquid-liquid extraction, gas absorption in dilute solutions, and leaching, the heat transfer rate is not as important. [1]

Humidification is one operation in which the mass transfer rate is affected by both heat and mass transfer. It involves the removal of water in the liquid phase to a gas mixture of air and water vapor. Water particles that come into contact with unsaturated air evaporate and are absorbed by the air stream. This reduces the temperature of the water due to heat being removed by the evaporated water particles. The most important type of humidification process is water cooling, in which warm water and air flow in countercurrent directions. Typically, the water flows down a column and air flows up the column, and close contact between the two is critical for obtaining large mass and heat transfer rates. [2]

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Industry applications of humidification involve controlling of the moisture content of air, such as in drying processes and air conditioning, and the cooling and recovery of water. [2,1] When controlling air humidity is critical, such as for indoor air quality or creating an extremely dry atmosphere, air conditioning and drying can be used. [2, 3] In operations where water is recovered, the cooled water is reused in heat exchangers to cool process streams. Also, cooled water is applied as a solution to cooling the surface of heat exchangers. [1]

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In this experiment, humidification is applied to study mass and heat transfer in a wetted wall column by obtaining experimental data on the temperature and humidity of air and water flowing through a column at steady state. From this data, experimental values for the mass and heat transfer coefficients are calculated. Theoretical values are calculated using the Sherwood number correlation and then compared to the experimental values. The unit used in this lab was a wetted humidification column made of plexiglass that is approximately 2 inches in diameter and 4 feet long. Water flowed down the sides of the column in order to wet the surface, and air flowed in a countercurrent direction. [4]

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Theory

Mass and heat transfer are described in terms of their transport coefficients, k_c, for mass, and h, for heat. The following flux equations detail the relationship between the coefficients and the flux:

$$q = h \cdot \Delta T \tag{1}$$

$$N_A = k_c \cdot \Delta c_A \tag{2}$$

For the heat flux equation, q is the flux and ΔT is the change in temperature, and for the mass flux equation, N_A is the flux and Δc_A is the change in concentration. In both of these equations, the coefficients are equal to the ratio of flux to a driving force. [4]

To obtain mass and heat transfer coefficients without the flux, correlations, in terms of dimensionless groups, are required. For the mass transfer coefficient of the gas phase in a circular pipe, the Sherwood number is correlated to the Reynolds and Schmidt numbers. [4]

$$Sh = 0.023 \cdot Re^{0.83} \cdot Sc^{0.44} \tag{3}$$

The Schmidt number, Sc, is the ratio of kinematic viscosity, ν, to the diffusion coefficient of the fluid, D_{AB}, and the Reynolds number is equal to the diameter, D, and fluid velocity, v, over the kinematic viscosity. [2, 5] Inserting the Sherwood number equation (6) into equation (3) allows for an equivalent equation that is used to solve for the mass transfer coefficient. [4]

$$Sc = \nu / D_{AB} \tag{4}$$

$$Re = D\mathrm{v}/\nu \tag{5}$$

$$Sh = k_c \cdot L / D_{AB} \tag{6}$$

L = length of the system under study

Mass and heat transfer occur simultaneously but do not always affect each other equally. Generally, heat transfer does not affect the mass transfer coefficient. Conversely, the heat transfer coefficient is influenced by mass transfer because the component diffusing into a fluid carries energy that can change the thermal conductivity of the fluid. In humidification systems, the correction factor accounting for this change is small and can be neglected. This is true when the mass flux and concentration of the diffusing component are small. Thus, the aforementioned correlations can be used to calculate the mass and heat transfer coefficients, which can then be used to calculate fluxes from equations (1) and (2). [4]

Transport coefficients describing heat and mass in a column depend on the differential of the average cross-sectional temperature and composition of the fluid. These changes are represented by a mass balance on the water. [4]

$$d(G_y) = k_c \cdot c(y_i - y)dA \tag{7}$$

The molar flowrate of the water in the gas across the differential column height, dG_y, is equal to the mass transfer rate, $k_c c(y_i - y)$, at a specified interface area, dA. The mol fractions of water at the interface and in the gas are represented by y_i and y, and c is the

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Units and assumptions. Note that some of the empirical equations only valid in a specific unit system, like CGS, or FPS.

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gas molar concentration. This relationship is also exhibited in Figure 1 which shows the direction of gas flow over a designated cross-sectional area of a column. These equations are valid when the mass transfer rate and concentrations of the transferring species are low. [4]

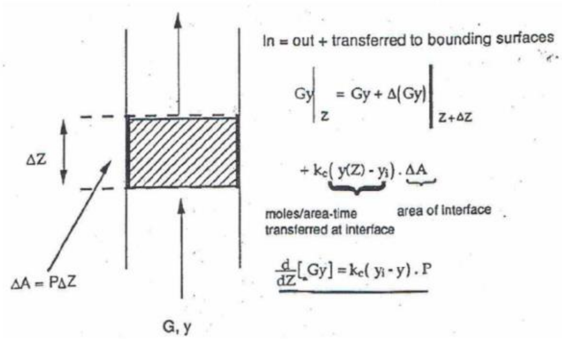

Figure 1: Schematic of a differential element in a column. [4]

In cases of high mass transfer rates and high concentrations of the transferring species, a bulk flow term is needed to correct for this high rate. The concentration of water vapor in the gas is expressed as a mass ratio, Y', of the mass of water to the mass of dry gas. This means the differential mass balance of water is rewritten in terms of mass, where G'_s is the mass flowrate of dry air, P is the perimeter of the column, and dz is the differential height. [4]

$$G_s' d(Y') = k_Y(Y_i' - Y')Pdz \tag{8}$$

The mass transfer coefficient is represented by k_y to define the coefficient in terms of the mass ratio of the concentration driving forces. Using the molecular weight of air, MW_{air}, the moles are converted to mass. To determine the best approximation of k_y, the log mean average over the length of the column, from inlet to outlet, $(1-y)_M$ is used. [4,6]

$$k_y = k_c(MW_{air})c(1-y)_M \tag{9}$$

Next, a heat balance describing the rate of heat transfer over a cross-sectional area is determined. [4]

$$d(GC_pT) = h(T_i - T)dA \tag{10}$$

Equation (10) is in terms of moles where C_p is the molar heat capacity of the gas, T_i is the interface temperature, and T is the gas temperature. Again, for humidification problems, a mass relationship is needed. The mass flow rate of dry air, G'_S, and humid heat or heat capacity of the gas, C'_S, are used. [4]

$$G_s' C_S' dT = h(T_i - T)dA \tag{11}$$

As long as the mass ratio, Y', does not vary much throughout the column, the humid heat is constant. Otherwise, the humid heat is the addition of the heat capacity of dry air and Y' times the heat capacity of water vapor. [4]

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Integration of equations (8) and (10) lead to the determination of the number of transfer units, NTU or N_tG, and the height of transfer unit, HTU. Both equations relate the inlet and outlet, subscripts 1 and 2, compositions and temperatures to the height. Y'_i and T_i are treated as constants. [4]

$$ln\left\{\frac{Y_i' - Y_1'}{Y_i' - Y_2'}\right\} = \frac{k_y P z}{G_S} = \text{NTU} \quad (12)$$

$$ln\left\{\frac{T_i' - T_1'}{T_i' - T_2'}\right\} = \frac{k_y P z}{\text{Cs} G_S} \quad (13)$$

NTU is dimensionless and describes the difficulty of separation as it provides a measure of the number of equilibrium stages required to reach a degree of separation. A large number is normally required for high product purity. HTU is of unit length and measures the separation efficiency of the separation process. The smaller the HTU, the larger the value of the mass transfer coefficient and the higher the efficiency of the mass transfer. [7]

Results

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You need to include the graphs that HTU vs. Flow Rate and RH% vs. Flow Rate, which are also not showing in your tables.

For this experiment, the objective was to determine the overall heat and mass transfer coefficients of the wetted wall column of the humidification unit. Dry air (with a relative humidity of approximately 7%) was sent through the column at different flow rates. An anemometer was used to measure the relative humidity and temperature of the air that was coming out at the top of the unit.

The experiment also called for the procedure to be repeated with various temperatures by heating the incoming air with steam in a heat exchanger before entering the column. Three temperature ranges were chosen, with 22°C being *No Heat*, 51°C being *Medium Heat* and temperatures above 58°C to be *High Heat*.

1. **Overall Heat Transfer Coefficient *(h)***

Table 1: Data for Heat Transfer Coefficients, No Heat

No Heat										
SCFM	T (air in) 'C	Density (kg/m3)	m3/min	Mass Flow (kg/min)	Re	Sc	Sh	kc	Cs	h
3	22.22	1.195	0.0849	0.1014555	2,322.5	0.66	11.91435	2.73E-06	3.939	3.329
6	22.22	1.195	0.1698	0.202911	4,645.1	0.66	21.18035	4.85E-06	3.705	6.086
9	22.22	1.195	0.2547	0.3043665	6,967.6	0.66	29.6542	6.79E-06	3.182	10.459

Commented [DL16]: Better to convert the tables into figures.

Table 2: Data for Heat Transfer Coefficients, Medium Heat

Medium Heat										
SCFM	T (air in) 'C	Density (kg/m3)	m3/min	Mass Flow (kg/min)	Re	Sc	Sh	kc	Cs	h
3	51	1.089	0.085	0.0906	1,972.5	0.66	10.404	2.38E-06	3.768	4.127
6	51	1.089	0.170	0.1812	3,945.0	0.66	18.495	4.23E-06	2.901	5.800
9	51	1.089	0.255	0.2662	6,183.7	0.66	26.857	6.15E-06	2.650	4.311

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Table 3: Data for Heat Transfer Coefficients, High Heat

High Heat										
SCFM	T (air in) 'C	Density (kg/m3)	m3/min	Mass Flow (kg/min)	Re	Sc	Sh	kc	Cs	h
3	58	1.067	0.085	0.091	1,899.6	0.66	10.084	2.31E-06	3.939	4.152
6	60	1.06	0.170	0.181	3,760.9	0.66	17.775	4.07E-06	2.721	4.655
9	65	1.045	0.255	0.266	5,488.3	0.66	24.325	5.57E-06	2.340	4.361

Equation 13 was used to calculate the values of h based off the experimental data obtained in tables 1-3. The Schmidt number was found from literature values for an air-water system [8] which meant that the Sherwood number could be calculated as well. Figure 2 shows the experimental heat transfer coefficient as a function of the calculated Reynolds numbers.

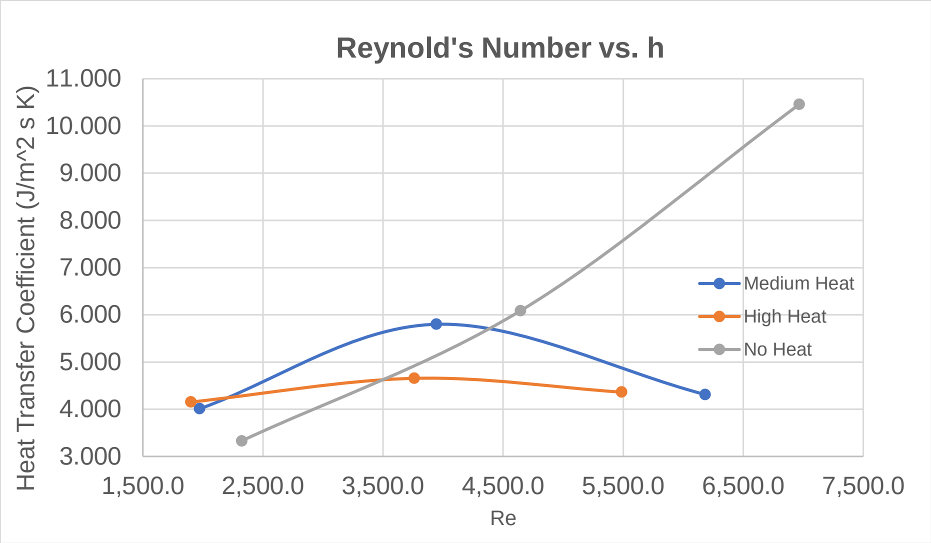

Figure 2: Relationship between Reynolds Number and Heat Transfer Coefficient ($J/m^2s.K$)

Additionally, the Schmidt number raised to 0.44 was divided by the Sherwood number and plotted against the Reynolds number on a logarithmic scale seen in figure 3.

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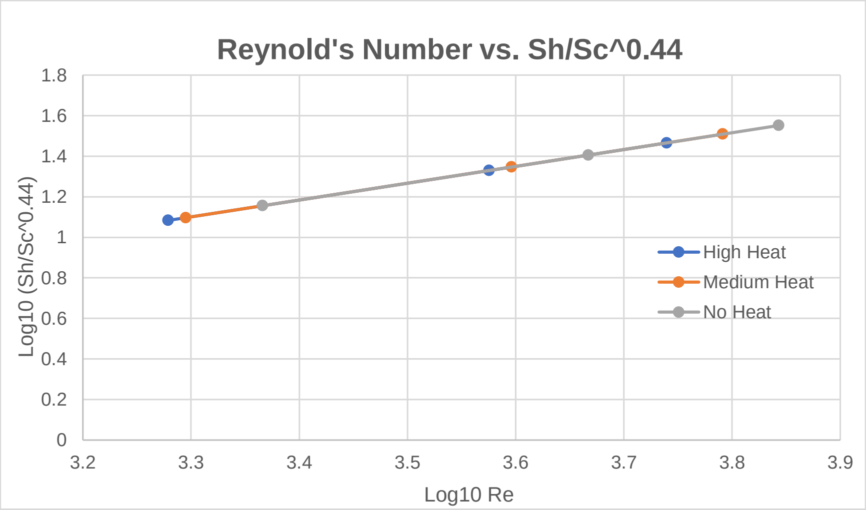

Figure 3: Graph of $\log_{10}$Re vs. $\log_{10}$ ($Sh/Sc^{0.44}$).

2. Overall Mass Transfer Coefficient

Table 4: Data for Mass Transfer Coefficients, No Heat

mole fracs of air	Density of air at temp measured (kg/m3)	Molar Density, Air	Absolute Humidity	Molar Abs. Humidity	total density	air frac	mol frac h20	mass frac h20	yi	ky
flow rate 1 in	1.195	0.041249569	0.0015	8.32639E-05	0.04133283	0.997986	0.002014475	0.00125523	0.991551853	0.009405562
flow rate 1 out	1.179	0.040697273	0.0184	0.001021371	0.04171864	0.975518	0.024482365	0.015606446		
flow rate 2 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.992712796	0.015788287
flow rate 2 out	1.181	0.04076631	0.0157	0.000871496	0.04163781	0.97907	0.020930401	0.013293819		
flow rate 3 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.994251006	0.017666189
flow rate 3 out	1.183	0.040835347	0.0121	0.000671663	0.04150701	0.983818	0.016181905	0.010228233		

Table 5: Data for Mass Transfer Coefficients, Medium Heat

mole fracs of air	Density of air at temp measured (kg/m3)	Molar Density, Air	Absolute Humidity	Molar Abs. Humidity	total density	air frac	mol frac h20	mass frac h20	yi	ky
flow rate 1 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.990716284	0.006681226
flow rate 1 out	1.164	0.040179496	0.0201	0.001115737	0.04129523	0.972981	0.02701854	0.017268041		
flow rate 4 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.99265532	0.010101435
flow rate 4 out	1.156	0.039903348	0.0155	0.000860394	0.04076374	0.978893	0.021106848	0.013408304		
flow rate 5 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.992453039	0.015660932
flow rate 5 out	1.144	0.039489127	0.0158	0.000877047	0.04036617	0.978273	0.021727274	0.013811189		

Table 6: Data for Mass Transfer Coefficients, High Heat

mole fracs of air	Density of air at temp measured (kg/m3)	Molar Density, Air	Absolute Humidity	Molar Abs. Humidity	total density	air frac	mol frac h20	mass frac h20	yi	ky
flow rate 2 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.992383623	0.005278537
flow rate 2 out	1.147	0.039592682	0.016	0.000888149	0.04048083	0.97806	0.021939984	0.013949433		
flow rate 3 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.993166389	0.009245506
flow rate 3 out	1.138	0.039282016	0.0141	0.000782681	0.0400647	0.980465	0.01953543	0.012390158		
flow rate 6 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.989324896	0.023573651
flow rate 6 out	1.158	0.039972385	0.0232	0.001287816	0.0412602	0.968788	0.031212056	0.020034542		

Equation 12 was used to calculate the mass transfer coefficient, ky, based off the experimental data obtained in tables 4 – 6. The mass fractions of water where obtained through the use of the measured relative humidity which was converted to absolute humidity. Figure 3 shows the mass transfer coefficient as a function of the calculated Reynolds number.

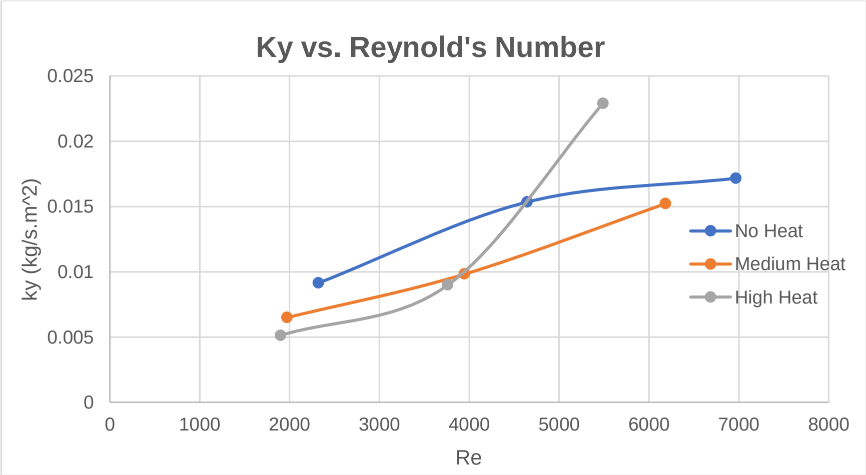

Figure 4: Graph of Reynolds Number vs. Mass Transfer Coefficient k_y ($Kg/s.m^2$)

Figure 4 shows the NTU as a function of the inlet temperature with airflow as a parameter. The NTU value is defined as the number of transfer units and is calculated using equation 12.

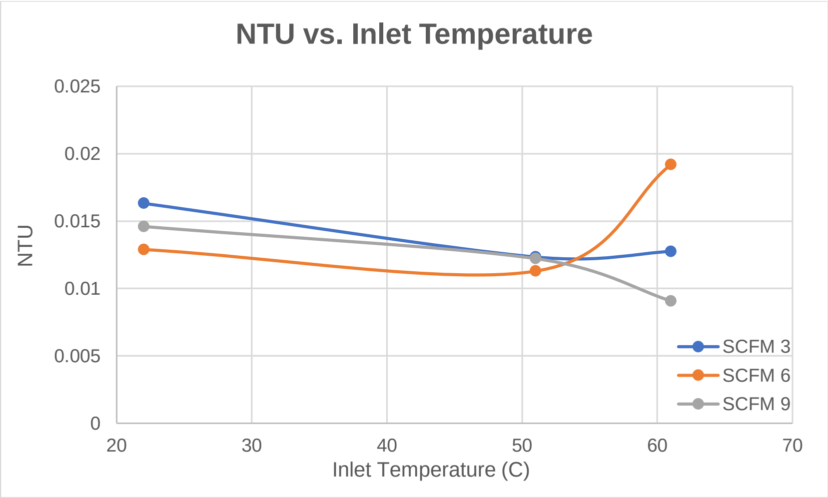

Figure 5: Graph of NTU vs. Inlet Temperature (^oC)

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Figure 5 shows the relationship of the measured mass transfer coefficient values, ky, as a function of the Sherwood numbers. The Sherwood numbers were calculated using the correlations given by equation 3.

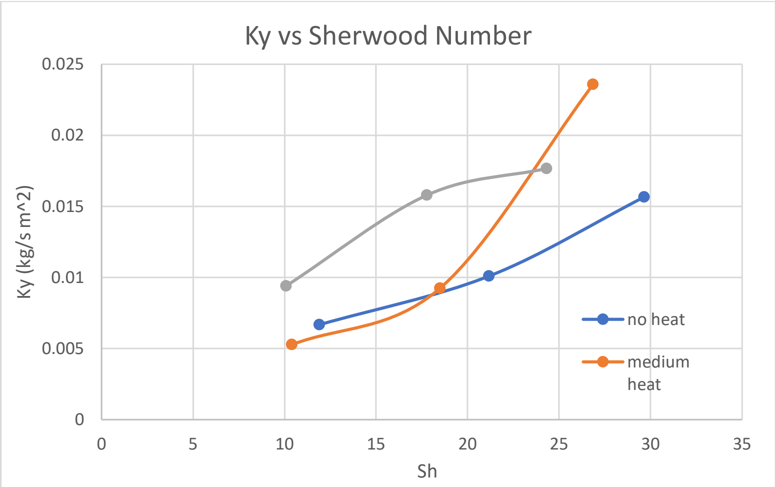

Figure 6: Graph of Sherwood Number vs. Mass Transfer Coefficient k_y (Kg/s.m^2)

Discussion

The objective of this experiment is to study heat and mass transport in a wetted wall column by comparing the values obtained from experimental methods to theoretical values calculated using known correlations for humidification.

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Where's the discussion about the humidity changes with parameters like the air flow rates?

When analyzing the mass transport of the wetted wall system the mass fraction of water obtained from the humidity of the system was used to find the experimental value of the mass transport coefficient. Figure 4 below shows the relationship between the experimental values of the mass transfer coefficient for three separate flowrates at three different temperature ranges. The trend shows that the mass transfer coefficient increases as the flowrate increases. The figure also shows that increasing the temperature has varying degrees of influence on the mass transfer coefficient. For the no heat condition, the mass transfer coefficient shows an almost asymptotic trend as flowrate increases while high heat conditions showed a sharp increase in mass transport as flowrate increased. This can be justified since the heated gas flowing through the column will likely cause more water to change phase from liquid to vapor thus increasing the mass flow rate.

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Commented [DL22]: Could be a reason. Also, the convective transport (driving forces) of the fluid could be considered.

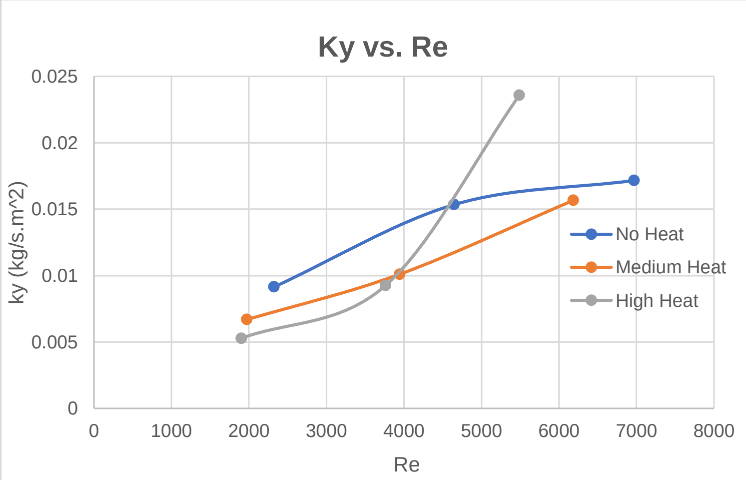

Figure 4: Graph of Reynolds Number vs. Mass Transfer Coefficient k_y ($Kg/s.m^2$)

When comparing figure 4 trends to those of figure 3 below it is seen that the overall trend of increasing flowrate is observed but the increase in temperature trend is not. This could be caused by several errors observed during the lab such as the inability to maintain constant air temperature for a suitable period of time for the humidity to stabilize. This can cause a discrepancy that the theoretical correlation did not account for in the humidification experiment.

Commented [DL23]: There could be. But more important things that you need to analyze the differences between the theoretical and the experimental values and present a discussion.

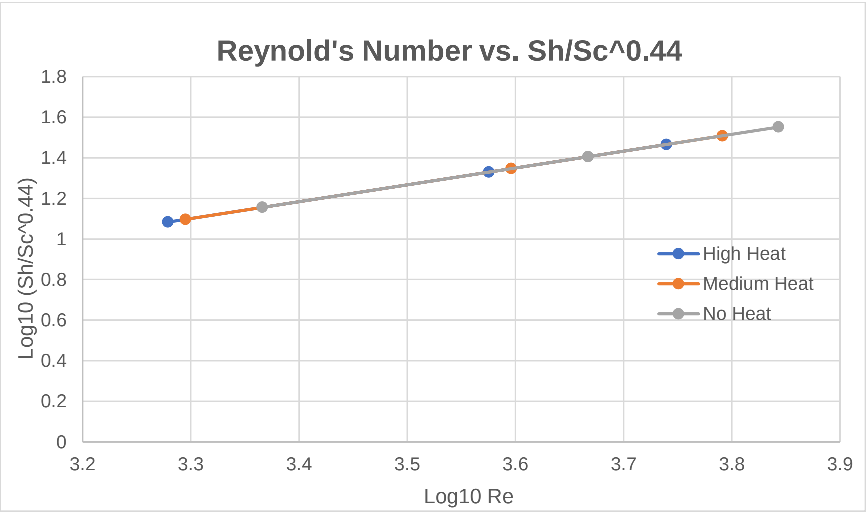

Figure 3: Graph of $\log_{10}$Re vs. $\log_{10}$ ($Sh/Sc^{0.44}$).

Another parameter observed for the mass transport portion of the data was the NTU values dependence on air temperature. NTU's are the number of transfer units which is a measurement of equilibrium stages needed to achieve a specific mass transfer. As the temperature of the air flowing through the column increases it is expected that the number of equilibrium stages needed will decrease due to the higher mass transfer rate. Figure 5 below shows a decreasing trend up to a certain temperature of about 53℃ where all flowrates deviate from the trend. This deviation may be caused by the unsteady state humidity readings at the higher temperature ranges.

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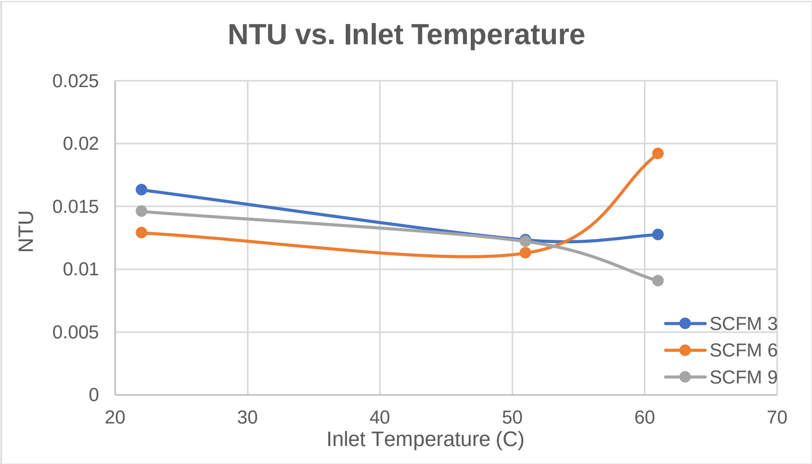

Figure 5: Graph of NTU vs. Inlet Temperature (°C)

Next, we analyzed the data in terms of heat transfer coefficients and compared it to the Sherwood correlation. Figure 2 below shows that the heat transfer coefficient increased only when there was unheated air flowing through the system at high flowrates while the other temperatures exhibited a lower heat transfer rate. It was explained earlier that the heat transfer coefficient can be influenced by the mass transfer of the system due to the diffusing component carrying energy that can change the thermal conductivity of the fluid. By this logic the heat transfer rate should increase but experimental data shows otherwise. There was a large error made while conducting the experiment at elevated gas temperatures that may account for this discrepancy as well at the lack of an equilibrium being reached between the air and the wetted column before measurements were taken. In comparison with figure 3, the increasing trend is only observed by the non-heated air.

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Commented [DL26]: Good discussion.

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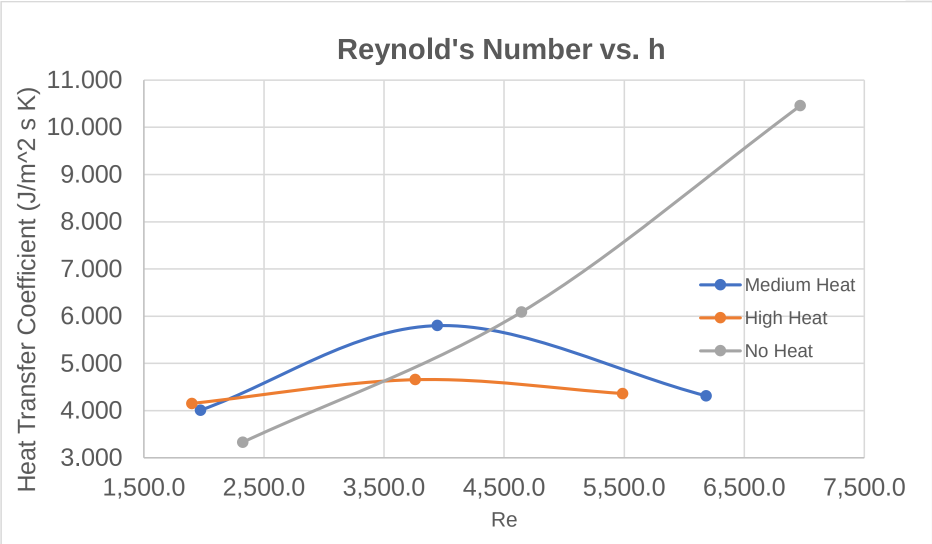

Figure 2: Relationship between Reynolds Number and Heat Transfer Coefficient ($J/m^2s.K$)

Overall the comparison between experimental heat and mass transfer coefficients and the theoretical values found using the Sherwood number were only marginally agreeable. The heating of the air in the experiment was the hardest variable to control and arguably the most important since the humidity and temperature data collected were collected based on time intervals rather than being collected during periods of stability. The temperature of the water being recycled would also change temperature since the temperature climbed higher than anticipated and data had to be collected from high to low temperatures for some flowrates. In the end our data showed that the Lewis number did not equal 1 with a large percent error over 100%.

Commented [DL28]: Usually happens for heat transfer experiments!

References

[1] “Simultaneous Heat and Mass Transfer I: Humidification.” Principles of Unit Operations. Chapter 17.

[2] Geankoplis, C.J., Hersel, A.A., and Lepek, D.H. “Humidification Processes.” Transport Processes and Separation Process Principles. Chapter 23.

[3] https://www.condairgroup.com/desiccant-dryers

[4] Zdunek, A. Humidification. CHE 382: Unit Operations Lab Manual.

[5] Chapter 3: Convective Mass Transfer Handout

[6] Chapter 8: Fundatmental of Mass Transfer Handout

[7] “Packing Height: The Method of Transfer Units.” http://www.separationprocesses.com/Absorption/GA_Chp04c.htm

[8] “Mass Diffusivity Data”, Martinez. Accessed from http://webserver.dmt.upm.es/~isidoro/dat1/Mass%20diffusivity%20data.pdf

[9] “Air- Dynamic and Kinematic Viscosity”, The Engineering Toolbox. https://www.engineeringtoolbox.com/air-absolute-kinematic-viscosity-d_601.html

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[3], [5] and [6] are not found in the report.
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Appendix I: Data Tabulation/Graphs

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Experimental Data:

Table 1: Data for Heat Transfer Coefficients, No Heat

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No Heat										
SCFM	T (air in) 'C	Density (kg/m3)	m3/min	Mass Flow (kg/min)	Re	Sc	Sh	kc	Cs	h
3	22.22	1.195	0.0849	0.1014555	2,322.5	0.66	11.91435	2.73E-06	3.939	3.329
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Table 2: Data for Heat Transfer Coefficients, Medium Heat

Medium Heat										
SCFM	T (air in) 'C	Density (kg/m3)	m3/min	Mass Flow (kg/min)	Re	Sc	Sh	kc	Cs	h
3	51	1.089	0.085	0.0906	1,972.5	0.66	10.404	2.38E-06	3.768	4.127
6	51	1.089	0.170	0.1812	3,945.0	0.66	18.495	4.23E-06	2.901	5.800
9	51	1.089	0.255	0.2662	6,183.7	0.66	26.857	6.15E-06	2.650	4.311

Table 3: Data for Heat Transfer Coefficients, High Heat

High Heat										
SCFM	T (air in) 'C	Density (kg/m3)	m3/min	Mass Flow (kg/min)	Re	Sc	Sh	kc	Cs	h
3	58	1.067	0.085	0.091	1,899.6	0.66	10.084	2.31E-06	3.939	4.152
6	60	1.06	0.170	0.181	3,760.9	0.66	17.775	4.07E-06	2.721	4.655
9	65	1.045	0.255	0.266	5,488.3	0.66	24.325	5.57E-06	2.340	4.361

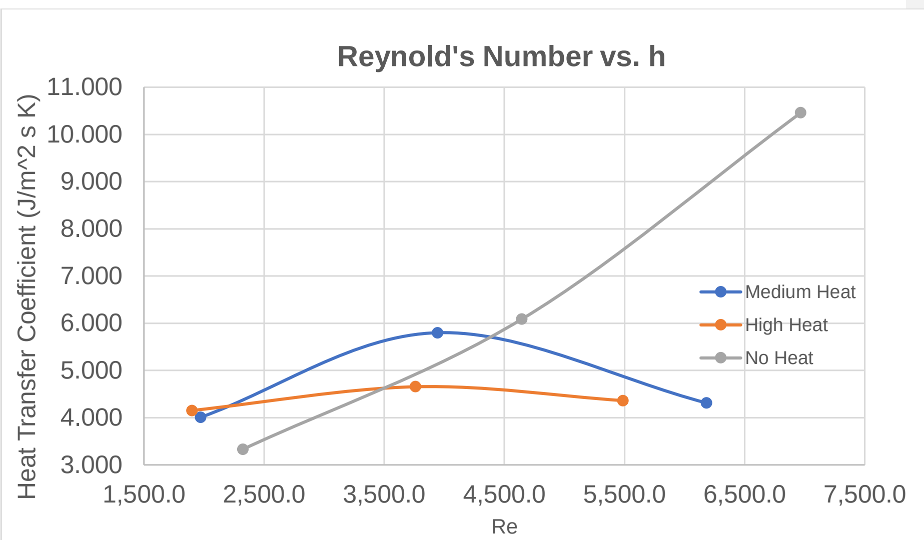

Figure 2: Relationship between Reynolds Number and Heat Transfer Coefficient ($J/m^2.s.K$)

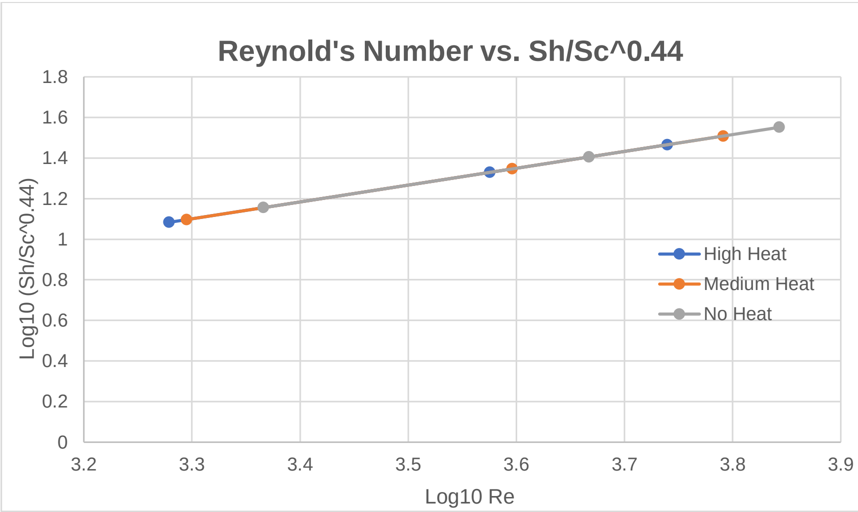

Figure 3: Graph of $\log_{10}$Re vs. $\log_{10}$ ($Sh/Sc^{0.44}$).

Table 4: Data for Mass Transfer Coefficients, No Heat

mole fracs of air	Density of air at temp measured (kg/m3)	Molar Density, Air	Absolute Humidity	Molar Abs. Humidity	total density	air frac	mol frac h20	mass frac h20	yi	ky
flow rate 1 in	1.195	0.041249569	0.0015	8.32639E-05	0.04133283	0.997986	0.002014475	0.00125523	0.991551853	0.009405562
flow rate 1 out	1.179	0.040697273	0.0184	0.001021371	0.04171864	0.975518	0.024482365	0.015606446		
flow rate 2 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.992712796	0.015788287
flow rate 2 out	1.181	0.04076631	0.0157	0.000871496	0.04163781	0.97907	0.020930401	0.013293819		
flow rate 3 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.994251006	0.017666189
flow rate 3 out	1.183	0.040835347	0.0121	0.000671663	0.04150701	0.983818	0.016181905	0.010228233		

Commented [DL35]: Units.

Table 5: Data for Mass Transfer Coefficients, Medium Heat

mole fracs of air	Density of air at temp measured (kg/m3)	Molar Density, Air	Absolute Humidity	Molar Abs. Humidity	total density	air frac	mol frac h20	mass frac h20	yi	ky
flow rate 1 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.990716284	0.006681226
flow rate 1 out	1.164	0.040179496	0.0201	0.001115737	0.04129523	0.972981	0.02701854	0.017268041		
flow rate 4 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.99265532	0.010101435
flow rate 4 out	1.156	0.039903348	0.0155	0.000860394	0.04076374	0.978893	0.021106848	0.013408304		
flow rate 5 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.992453039	0.015660932
flow rate 5 out	1.144	0.039489127	0.0158	0.000877047	0.04036617	0.978273	0.021727274	0.013811189		

Commented [DL36]: Significant numbers.

Table 6: Data for Mass Transfer Coefficients, High Heat

mole fracs of air	Density of air at temp measured (kg/m3)	Molar Density, Air	Absolute Humidity	Molar Abs. Humidity	total density	air frac	mol frac h20	mass frac h20	yi	ky
flow rate 2 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.992383623	0.005278537
flow rate 2 out	1.147	0.039592682	0.016	0.000888149	0.04048083	0.97806	0.021939984	0.013949433		
flow rate 3 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.993166389	0.009245506
flow rate 3 out	1.138	0.039282016	0.0141	0.000782681	0.0400647	0.980465	0.01953543	0.012390158		
flow rate 6 in	1.194	0.04121505	0.0015	8.32627E-05	0.04129831	0.997984	0.002016127	0.001256262	0.989324896	0.023573651
flow rate 6 out	1.158	0.039972385	0.0232	0.001287816	0.0412602	0.968788	0.031212056	0.020034542		

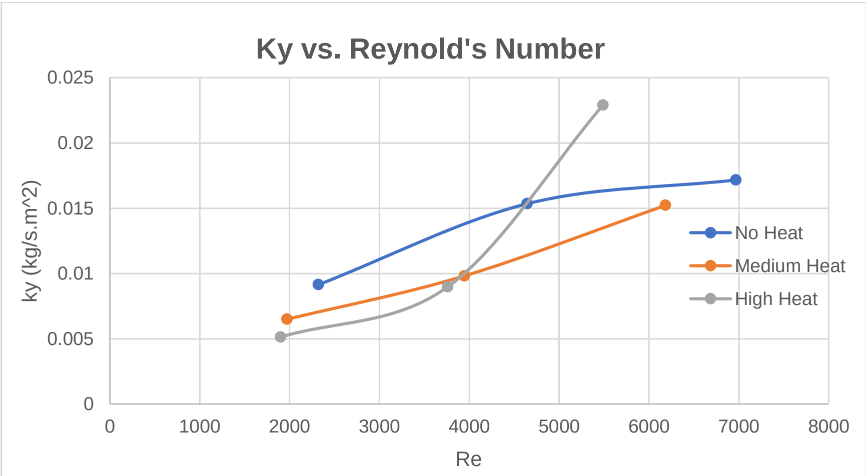

Figure 4: Graph of Reynolds Number vs. Mass Transfer Coefficient k_y ($Kg/s.m^2$)

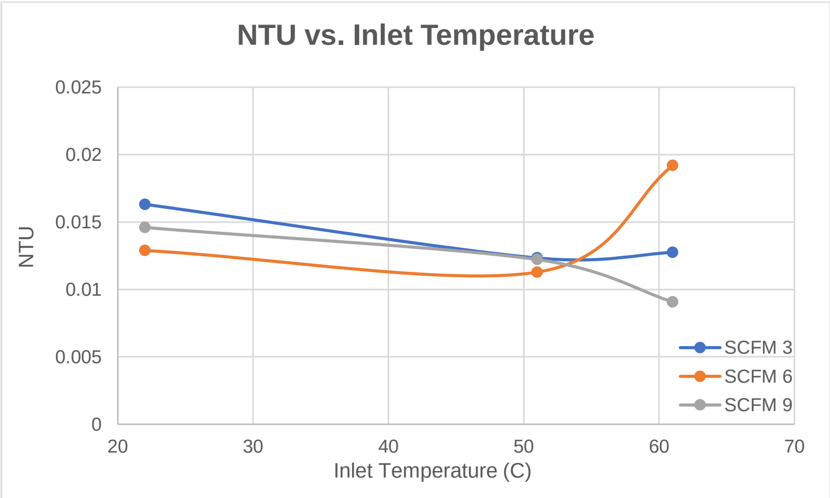

Figure 5: Graph of NTU vs. Inlet Temperature (^oC)

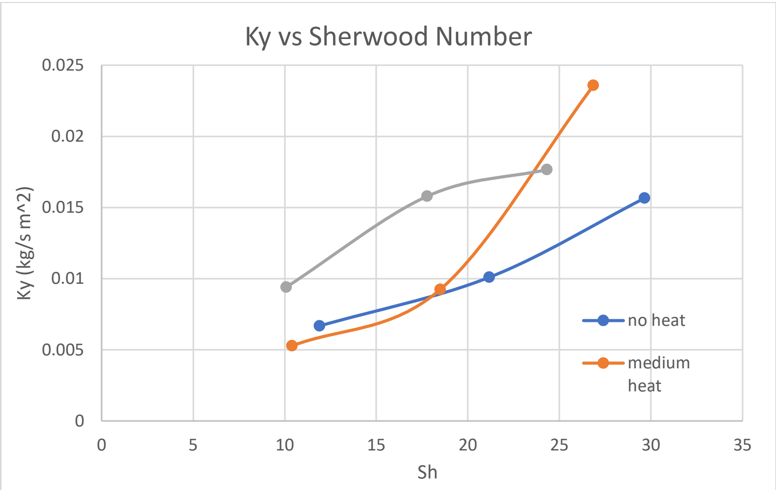

Figure 6: Graph of Sherwood Number vs. Mass Transfer Coefficient k_y (Kg/s.m^2)

Table 7: Raw Data Collected from Experiment, No Heat, 3 SCFM

Flowrate 1	Units	Value		
Flowrate of water used to wet wall:	GPM	0.2282		
Flowrate unheated air:	SCFM	3		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	32	31	31
Temp water in reservoir T6:	C	30		
Temp of unheated air in:	F	72		
Temp of unheated air out	C	26.1		
Humidity S.State unheated air:	%	75		

Table 8: Raw Data Collected from Experiment, No Heat, 6 SCFM

Flowrate 2	Units	Value		
Flowrate of water used to wet wall:	GPM	0.24124		
Flowrate unheated air:	SCFM	6		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	31	31	29
Temp water in reservoir T6:	C	29		
Temp of unheated air in:	F	72		
Temp of unheated air out	C	25.8		
Humidity S.State unheated air:	%	64.5		

Table 9: Raw Data Collected from Experiment, No Heat, 9 SCFM

Flowrate 3	Units	Value		
Flowrate of water used to wet wall:	GPM	0.24124		
Flowrate unheated air:	SCFM	9		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	29	28	27
Temp water in reservoir T6:	C	27		
Temp of unheated air in:	F	72		
Temp of unheated air out	C	25.2		
Humidity S.State unheated air:	%	52		

Table 10: Raw Data Collected from Experiment, Medium Heat, 3 SCFM

Flowrate 1	Units	Value		
Flowrate of water used to wet wall:	m3/s	1.52199E-05		
Flowrate heated air:	SCFM	3		
Temperature of heated air in:	C	51		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	32	31	31
Temp water in reservoir T6:	C	31		
Temp of heated air out:	C	30		
Humidity S.State heated air:	%	66		
Temperature of water into col	C	32		

Table 11: Raw Data Collected from Experiment, Medium Heat, 6 SCFM

Flowrate 2	Units	Value		
Flowrate of water used to wet wall:	m3/s	1.52199E-05		
Flowrate heated air:	SCFM	6		
Temperature of heated air in:	C	51		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	32	31	31
Temp water in reservoir T6:	C	31		
Temp of heated air out:	C	32.3		
Humidity S.State heated air:	%	45.3		
Temperature of water into col	C	32		

Table 12: Raw Data Collected from Experiment, Medium Heat, 9 SCFM

Flowrate 3	Units	Value		
Flowrate of water used to wet wall:	m3/s	1.52199E-05		
Flowrate heated air:	SCFM	9		
Temperature of heated air in:	C	51		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	32	31	31
Temp water in reservoir T6:	C	31		
Temp of heated air out:	C	35.4		
Humidity S.State heated air:	%	39.3		
Temperature of water into col	C	32		

Table 13: Raw Data Collected from Experiment, High Heat, 3 SCFM

Flowrate 1	Units	Value		
Flowrate of water used to wet wall:	m3/s	1.52199E-05		
Flowrate heated air:	SCFM	3		
Temperature of heated air in:	C	58		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	34	33	33
Temp water in reservoir T6:	C	33		
Temp of heated air out:	C	31.6		
Humidity S.State heated air:	%	70.1		
Temperature of water into col	C	34		

Table 14: Raw Data Collected from Experiment, High Heat, 6 SCFM

Flowrate 2	Units	Value		
Flowrate of water used to wet wall:	m3/s	1.52199E-05		
Flowrate heated air:	SCFM	6		
Temperature of heated air in:	C	60		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	33	32	32
Temp water in reservoir T6:	C	32		
Temp of heated air out:	C	34.7		
Humidity S.State heated air:	%	41		
Temp of water into col	C	33		

Table 15: Raw Data Collected from Experiment, High Heat, 9 SCFM

Flowrate 3	Units	Value		
Flowrate of water used to wet wall:	m3/s	1.52199E-05		
Flowrate heated air:	SCFM	9		
Temperature of heated air in:	C	65		
		Location 1 T3	Location 2 T4	Location 3 T5
Temp S.State water and air:	C	33	31	31
Temp water in reservoir T6:	C	31		
Temp of heated air out:	C	37		
Humidity S.State heated air:	%	31.9		
Temp of water in col:	C	33		

Appendix II: Error Analysis

Commented [DL37]: 9/10
Quantitative error analysis about the dimensionless groups' correlation and the uncertainty of the apparatuses are needed.

Humidification is a simple and cheap method for collecting experimental data in calculations of heat and mass transfer coefficients, by using wet and dry bulb temperature sensors. However, the error is often very high in this type of apparatus due to the relative humidity measurement uncertainty. To increase the accuracy, it is important to calibrate the temperature sensors, although the calibration can be very challenging. Proper handling and maintenance are also required, but there are many other factors that contribute to the low accuracy as wick being dry, dirty, not properly pooled etc.
Throughout the experiment multiple difficulties in achieving of the steady state occurred. One of the challenges was to maintain the temperature of the heated air constant, where the lowest temperature range achieved was between 58 to 65 degrees Celsius.
Furthermore, it was not possible to measure steady state humidity because the electronic hygrometer would turn off or the temperature would not remain constant.
Finally, during the higher temperatures experimental phase, water in the reservoir heated up measurements and calculations taken at the hotter water temperatures may differ from the calculations made at the lower water temperatures due to the changes in properties. This may account to the variations of the raw data of the heated apparatus, since it was difficult for the humidity to stabilize.

Appendix III: Sample Calculations

Commented [DL38]: 9/10

*All sample calculations are done for a flowrate of 3 SCFM and an incoming air temperature of 51℃.

1. **Reynolds number calculation**

$$Re = \frac{QD}{\nu A} = \frac{0.085\,\frac{m^3}{min} * 0.0508\,m}{17.98 * 10^{-6}\,\frac{m^2}{s} * 60\,\frac{s}{min} * 0.0015205\,m^2} = \mathbf{1972.5}$$

Where,
Q is volumetric flowrate. For 3 SCFM = 0.085 m^3/min
D is diameter of pipe. 2 in. = 0.0508 m
ν is kinematic viscosity. For 51℃, ν = 17.98 m^2/s x 10^{-6} [9]
A is cross sectional area of pipe. $A = \frac{\pi}{4}D^2 = 0.0015205\,m^2$

2. **Schmidt number calculation**

****** Value corresponds to temperature of 300K

$$Sc = \frac{\nu}{D_{H_2O/Air}} = \mathbf{0.66}$$ [8]

Where,
ν is kinematic viscosity
$D_{H2O/Air}$ is diffusivity for water and air

Commented [DL39]: Units.

3. **Sherwood number calculation**

$$Sh = 0.023 * Re^{0.83} * Sc^{0.44}$$
$$Sh = 0.023 * 1972.5^{0.83} * 0.66^{0.44} = \mathbf{10.404}$$

4. **Perimeter of tube, P, conversion**

$$P = 8\,in. * \frac{0.0254m}{1\,in.} = \mathbf{0.2032\,m}$$

Where,
P is the circumference of the tube measured to be 8 inches

5. **Height of tube, z, conversion**

$$z = 4\,ft * \frac{0.3048m}{1\,ft} = \mathbf{1.2192\,m}$$

Where,
Z is the height of the tube measured to be 4 feet

6. Mass flowrate for gas, G_s, calculation

$$G_{s_{Heat}} = \rho\dot{V} = 1.089\frac{kg}{m^3} * 0.085\frac{m^3}{min} = \mathbf{0.09}\frac{\boldsymbol{kg}}{\boldsymbol{min}}$$

$$G_{s_{NoHeat}} = \rho\dot{V} = 1.195\frac{kg}{m^3} * 0.085\frac{m^3}{min} = \mathbf{0.010}\frac{\boldsymbol{kg}}{\boldsymbol{min}}$$

Where,
ρ is the density of the gas. 1.089 kg/m^3
$\dot{V}$ is the volumetric flowrate. 3 SCFM = 0.085 m^3/min

7. Air Fraction and mole fraction of water

$$\rho_{humidity} = \frac{AbsHumidity}{MW_{H_2O}} = 0.001157$$

$$\rho_{total} = \rho_{air} + \rho_{humidity} = 0.04129$$

$$F_{air} = \frac{\rho_{air}}{\rho_{total}} = 0.973$$

$$y_2 = 1 - 0.973 = \mathbf{0.027}$$

Commented [DL40]: Units.

8. Mass fraction of water, Y', calculation

$$Y_1' = \left(\frac{y}{1-y}\right)\left(\frac{MW_{H_2O}}{MW_{Air}}\right) = \left(\frac{0.00202}{1-0.00202}\right)\left(\frac{18.015}{28.97}\right) = \mathbf{0.00126}$$

$$Y_2' = \left(\frac{y_2}{1-y_2}\right)\left(\frac{MW_{H_2O}}{MW_{Air}}\right) = \left(\frac{0.027}{1-0.027}\right)\left(\frac{18.015}{28.97}\right) = \mathbf{0.0173}$$

$$Y_i' = \frac{Y_1' - Y_2'}{\ln\left(\frac{1-Y_2'}{1-Y_1'}\right)} = \frac{0.00126 - 0.0173}{\ln\left(\frac{1-0.0173}{1-0.00126}\right)} = \mathbf{0.9907}$$

Where,
Y is the mol fraction of water
MW_{H2O} is the molecular weight of water
MW_{air} is the molecular weight of air

9. Mass transfer coefficient, k_y, calculation

$$k_y = \frac{G_s}{Pz}\ln\left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'}\right) = \frac{0.01\frac{kg}{min}}{0.2032m * 1.2192m}\ln\left(\frac{0.99 - 0.00126}{0.99 - 0.01723}\right) = \mathbf{0.00668}\frac{\boldsymbol{kg}}{\mathbf{min}\,\boldsymbol{m^2}}$$

Where,
G_s is the mass flowrate for the unheated gas
P is the perimeter of the column
Z is the height of the column
Y_i' is the mass fraction of water in the column
Y_1' is the inlet composition of the gas
Y_2' is outlet composition of the gas

10. Number of transfer units, N_{tG}, calculation

$$N_{tG} = \ln\left(\frac{Y_i' - Y_1'}{Y_i' - Y_2'}\right) = \ln\left(\frac{0.99 - 0.00126}{0.99 - 0.01723}\right) = \mathbf{0.0163}$$

11. Humid heat capacity, C_s, calculation

$$C_s = C_{air} + Y'C_{H_2O} = 1.005\ \frac{m^2}{min^2℃} + 0.66 * 4.186\ \frac{m^2}{min^2℃} = \mathbf{3.768}\,\frac{\boldsymbol{m^2}}{\boldsymbol{min^2}℃}$$

Where,
C_{air} is the heat capacity of the dry air
Y' is the vapor content of the gas
C_{H2O} is the heat capacity of the water vapor

12. Heat transfer coefficient, h, calculation

$$h = \frac{G_sC_s}{Pz}\ln\left(\frac{T_i - T_1}{T_i - T_2}\right) = \frac{0.09\,\frac{kg}{min} * 3.768\,\frac{kg\ m^2}{\mathrm{min}^2℃}}{0.2032\ m * 1.2192m}\ln\left(\frac{31℃ - 51℃}{31℃ - 30℃}\right) = \mathbf{4.009}\,\frac{\boldsymbol{kg}}{\boldsymbol{min^3}℃}$$

Where,
G_s is the mass flowrate for the heated gas.
C_s is the humid heat capacity
P is the perimeter of the column
Z is the height of the column
T_i is the temperature at the interface
T_1 is the inlet temperature of the vapor
T_2 is the outlet temperature of the vapor

13. Lewis relationship calculation

$$Le = \frac{h}{k_yC_s} = \frac{4.009}{0.00668 * 3.768} = 159.275 \neq 1$$

Appendix IV: Individual Team Contributions

NAME: Alice Guzik

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)	0.5	
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	2	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	2	
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)	7	
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	2	
Total	21.5	

NAME: Adisa Hadzic

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)		
Apparatus (prelab)	1	
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing		
Executive Summary (final lab)	4	
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)		
Data Tabulation/Graphs (final)		
Error Analysis (final lab)	2	
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	14.5	

NAME: Kellen Krupa

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	0.25	
Experimental Objective (Prelab)		
Apparatus (prelab)	0.5	
Materials & Supplies (prelab)		
Procedure (prelab)	1	
Project Safety Evaluation (prelab)		
Final Lab Editing	0.25	
Executive Summary (final lab)		
Introduction (Final lab)	4	
Theory (Final Lab)	5	
Results (final lab)		
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	0.25	
Data Tabulation/Graphs (final)		
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	14.75	

NAME: Ian Lim

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	6	
Pre-Lab Editing	1	
Experimental Objective (Prelab)		
Apparatus (prelab)		
Materials & Supplies (prelab)		
Procedure (prelab)		
Project Safety Evaluation (prelab)		
Final Lab Editing	2	
Executive Summary (final lab)		
Introduction (Final lab)		
Theory (Final Lab)		
Results (final lab)	7	
Discussion (final lab)		
Conclusion (final lab)		
References (final and/or prelab)	1	
Data Tabulation/Graphs (final)	2	
Error Analysis (final lab)		
Sample Calculations (final Lab)		
Power Point Presentation	1	
Total	21	

NAME: Alexis Miranda

	TIME (HOURS)	DESCRIPTION
Operator (Both Lab Days)	3	
Pre-Lab Editing	0.5	
Experimental Objective (Prelab)	0	
Apparatus (prelab)	0.5	
Materials & Supplies (prelab)	0	
Procedure (prelab)	0	
Project Safety Evaluation (prelab)	1	
Final Lab Editing	1	
Executive Summary (final lab)	0.5	
Introduction (Final lab)	0	
Theory (Final Lab)	0	
Results (final lab)	0	
Discussion (final lab)	4	
Conclusion (final lab)	0	
References (final and/or prelab)	0.1	
Data Tabulation/Graphs (final)	0	
Error Analysis (final lab)	0	
Sample Calculations (final Lab)	3	
Power Point Presentation	1	
Total	14.6	